Notes: Ozdagli and Weber (RFS, 2026)

Monetary Policy through Production Networks: Evidence from the Stock Market

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1 Big picture

Question. How much of the stock market response to monetary policy shocks is transmitted through production networks rather than through only the direct exposure of each industry to aggregate demand?

Setting. The paper uses industry stock returns around scheduled FOMC announcements from 1994–2008, federal funds futures surprises as monetary policy shocks, and U.S. Bureau of Economic Analysis input-output tables to measure supplier-customer linkages.

Core challenge. A monetary shock hits the whole economy at once. If all industries move together, it is hard to say whether this is because every industry reacts directly to the shock or because shocks propagate through input-output chains.

Main idea. Treat the production network as a spatial-weighting matrix. A simple model with intermediate inputs implies that industry returns should satisfy a spatial autoregression: returns respond to the policy shock directly and also to a network-weighted average of other industry returns.

Main conceptual contribution. The paper gives a structural interpretation to a spatial autoregression in a production-network setting. The input-output matrix is not an ad hoc spatial matrix; it comes from the model's intermediate-input structure.

Main empirical design. The authors estimate industry-level event-window returns on:

- the federal-funds-futures monetary policy surprise;
- a BEA input-output weighted average of other industry returns;
- specifications with constant and heterogeneous industry coefficients.

Main baseline result. A 1 percentage point tighter-than-expected federal funds target-rate announcement lowers the aggregate stock market by about 3 percentage points. In the network decomposition, higher-order network effects account for roughly 55%–85% of the total response, depending on the decomposition.

Main mechanism. Monetary policy is interpreted as a demand shock. Expansionary policy raises final demand for consumer-facing industries. Those industries raise demand for inputs, which raises revenue for their suppliers, which then raises demand further upstream. Because the input-output network has cycles, the effect can feed back through multiple orders.

Key heterogeneity result. Industries close to end-consumers have much larger direct effects. Industries far from end-consumers have much larger network effects. This matches the upstream-propagation interpretation of a demand shock.

Key robustness result. The network result survives controls for Fama–French risk factors, alternative event types, subperiods, different industry aggregation levels, diagonal-entry restrictions, external-finance splits, and price-stickiness splits.

Economic takeaway. Production networks are not only relevant for idiosyncratic supply shocks. They are also a major transmission mechanism for aggregate monetary policy shocks.

2 Data and measurement (key objects)

2.1 Monetary policy shocks

The paper measures monetary policy surprises using federal funds futures around FOMC announcements. The shock is the change in the futures-implied federal funds rate from just before to just after the announcement, adjusted for the fact that futures settle on the monthly average federal funds rate:

$$v_t = \frac{D}{D-t}\left(ff^0_{t+1,t+} - ff^0_{t-1,t-}\right), \tag{1}$$

where D is the number of days in the month and t is the announcement date within the month.

Interpretation. v_t is the unexpected component of monetary policy. A positive value means policy is tighter than markets expected.

2.2 Event returns

The main outcome is industry stock returns in a narrow 30-minute window around scheduled FOMC press releases. The authors use NYSE TAQ transaction data, calculate firm-level returns around the window, and aggregate them into BEA industries using value weights.

Interpretation. Because the window is short, stock returns are intended to isolate the market's reassessment of future cash flows and discount rates caused by the monetary policy news, rather than by other macro shocks.

2.3 Input-output network

The production network is built from BEA benchmark input-output tables. The authors combine the BEA make table and use table to construct a supplier-by-customer matrix.

Let $MAKE$ be the make table and USE be the use table. The paper defines:

$$SHARE = MAKE \oslash (\mathbf{1} \times MAKE), \tag{2}$$

$$REVSHARE = SHARE \times USE, \tag{3}$$

$$SUPPSHARE = [REVSHARE \oslash (\mathbf{1} \times USE)]'. \tag{4}$$

The resulting matrix corresponds to the empirical input-output matrix W.

Interpretation. W_{ij} measures the share of industry i's intermediate inputs bought from supplier industry j. The transpose W' captures upstream propagation from customers to suppliers.

2.4 Industry aggregation

The paper uses BEA summary accounts and maps BEA industry codes to SIC and NAICS codes. The sample typically contains about 61–71 industries, depending on the mapping and event window.

Interpretation. The use of industries rather than firms makes linkages more stable and more technology-based. It also reduces concerns that firm-specific financial frictions dominate the results.

2.5 Sample period

The main sample runs from the first scheduled FOMC meeting in February 1994 through December 2008. The sample excludes the zero-lower-bound period and focuses on scheduled meetings.

Interpretation. Starting in 1994 is important because FOMC communication became clearer and event timing became less ambiguous.

2.6 Realized fundamentals

The paper also studies quarterly sales and operating income to see whether network effects appear in ex post fundamentals, not just in stock returns.

For sales, the authors construct:

$$\Delta sale_{i,t,H} = \frac{\frac{1}{4}\sum_{s=t+H}^{t+H+3} sale_{i,s} - \frac{1}{4}\sum_{s=t-4}^{t-1} sale_{i,s}}{TA_{i,t-1}} \times 100. \tag{5}$$

Interpretation. This compares four quarters after the policy shock with four quarters before the shock, scaled by assets, to reduce seasonality and size effects.

3 Models and empirical specifications (all equations + interpretation)

3.1 Static production-network model

Firm i chooses labor and intermediate inputs to maximize:

$$\pi_i = \max_{l_i, \{x_{ij}\}} p_i y_i - \sum_{j=1}^{N} p_j x_{ij} - w l_i - f_i, \tag{6}$$

with production function:

$$y_i = l_i^{\lambda_i} \left(\prod_{j=1}^{N} x_{ij}^{\omega_{ij}} \right)^{\alpha_i}. \tag{7}$$

The first-order conditions imply:

$$\alpha_i \omega_{ij} R_i = p_j x_{ij}, \tag{8}$$

$$\lambda_i R_i = w l_i, \tag{9}$$

where $R_i = p_i y_i$ is revenue.

Interpretation. The share of firm i's revenue spent on input j equals the intermediate-input share times the input weight. This is the microfoundation for using input-output shares as network weights.

3.2 Goods-market clearing and network transmission

Goods-market clearing leads to:

$$R_i = b_i \sum_{k=1}^{N} (1 - \alpha_k) R_k + \sum_{j=1}^{N} \alpha_j \omega_{ji} R_j. \tag{10}$$

In matrix form:

$$\left(I - W' D(\alpha)\right) R = bM. \tag{11}$$

Interpretation. Revenues depend on final consumer demand and on the demand of other industries for intermediate inputs. Since W' maps customers to suppliers, demand shocks travel upstream.

3.3 Monetary policy and profit response

With cash-in-advance for consumption goods, money supply pins down final expenditure. Log-linearizing profits gives:

$$\widehat{\pi} = \left(I - W' D(\alpha)\right)^{-1} \beta \widehat{M}. \tag{12}$$

Interpretation. The Leontief inverse converts the direct monetary shock into total effects after all higher-order network feedbacks.

3.4 Baseline spatial autoregression

The empirical specification is:

$$r_t = c + \beta v_t + W' D(\rho) r_t + \varepsilon_t. \tag{13}$$

Here:

- r_t is the vector of industry event-window returns;
- v_t is the monetary policy surprise;
- W is the BEA input-output matrix;
- $D(\rho)$ governs how strongly returns propagate through the production network;
- β captures direct industry exposure to the policy shock.

Interpretation. The regression asks whether an industry's return moves not only because of the policy surprise itself, but also because its customers' and suppliers' returns move through the input-output network.

3.5 Reduced-form inverse and higher-order effects

Solving the spatial autoregression gives:

$$r_t = \left(I - W'D(\rho)\right)^{-1} \beta v_t + \left(I - W'D(\rho)\right)^{-1} \varepsilon_t. \tag{14}$$

The inverse can be expanded as:

$$\left(I - W'D(\rho)\right)^{-1} = I + W'D(\rho) + \left(W'D(\rho)\right)^2 + \cdots . \tag{15}$$

Interpretation. The identity matrix is the direct effect. The first-order term is the effect through direct input-output links. Higher powers capture second-order, third-order, and further propagation through the network.

3.6 Three decompositions of total effects

The paper uses three ways to split total responses into direct and network components.

Decomposition 1: Pace–LeSage. Direct effects are the diagonal elements of the spatial multiplier; network effects are the off-diagonal elements.

Decomposition 2: Acemoglu–Akcigit–Kerr. Direct effects are the impact coefficients β; all higher-order terms in the Leontief expansion are network effects.

Decomposition 3: autarky comparison. The paper compares the empirical network economy with an autarkic economy in which each industry only uses its own output as an input. The difference is interpreted as network amplification.

Interpretation. The first two decompositions ask how much of the total response is direct versus propagated. The third asks how much the actual U.S. production network amplifies the response relative to an economy without cross-industry input-output links.

4 Identification and credibility

4.1 High-frequency monetary policy identification

The paper uses narrow event windows around FOMC announcements. Within this window, the monetary policy surprise is plausibly the main economy-wide shock affecting industry returns.

Credibility logic. This avoids the usual macro timing problem. Asset prices react within minutes, so the paper does not need quarterly or monthly exclusion restrictions.

4.2 Why stock returns are useful

Stock prices summarize expected future cash flows and discount rates. If monetary policy changes expected industry cash flows through production networks, this should be reflected in returns immediately.

Credibility logic. The stock market is used as a high-frequency laboratory for the real effects of monetary policy.

4.3 Why the input-output matrix identifies network effects

The BEA input-output matrix is measured before or independently of the specific FOMC event. Industry linkages change slowly and are mainly technological rather than chosen in response to a single policy announcement.

Credibility logic. The network matrix is not estimated from the same return comovement that the paper wants to explain; it comes from production data.

4.4 Main confounds and responses

Common factor comovement. Industry returns may comove because of market, size, or value factors. The paper controls for Fama–French factors and also uses factor-orthogonalized returns.

Mechanical network assignment. A concern is that any sparse matrix could produce large network effects. The paper uses Monte Carlo placebo networks and shows that random matrices with the same sparsity or outdegree distributions generally cannot replicate the empirical effects.

Diagonal entries. If industries buy heavily from themselves, within-industry effects might look like network propagation. The paper sets diagonal entries to zero and still finds large network effects.

Financial-friction channel. Monetary policy could affect firms through financing costs rather than demand. The paper splits industries by external finance dependence and the Kaplan–Zingales index and finds similar network shares.

Price-stickiness channel. The paper splits industries by price-adjustment frequency and finds similar network effects across groups.

4.5 What the paper can and cannot prove

The paper credibly shows that stock-market reactions to monetary surprises line up with the U.S. input-output network and that the implied network component is large.

It is more cautious about fully separating cash-flow news from discount-rate news. The fundamentals exercise helps, because similar network effects appear in sales and operating income, but the main high-frequency evidence is still based on stock returns.

5 Tables and figures

5.1 Figure 1: orders of the Leontief expansion

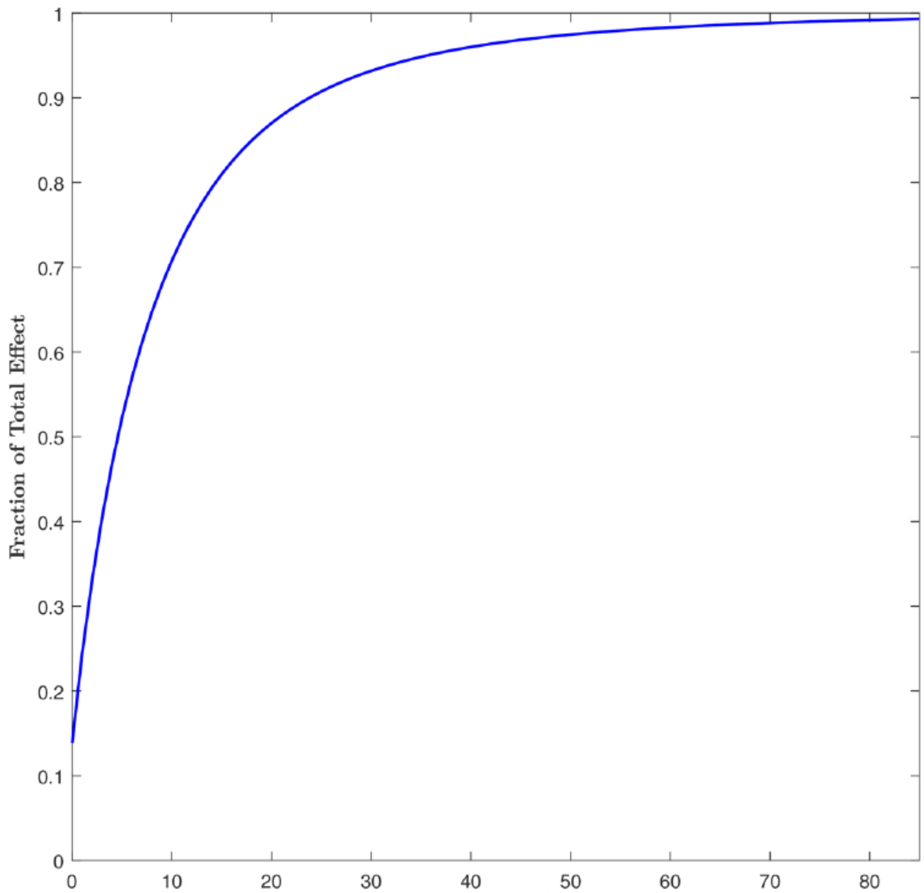


Figure 1. Fraction of the total effect captured as more orders of the Leontief expansion are added.

Read.

- The zero-order term captures only about 14% of the total stock-market response.
- The first 10 orders capture only about 71% of the total response.
- Many higher-order network paths are needed to recover the full effect.

Economic message. Monetary-policy transmission is not just direct customer-supplier pass-through. Long chains and feedback loops matter.

5.2 Table 1: constant-coefficient baseline

Table 1
Response of industry returns to monetary policy shocks: Constant coefficients

	OLS	OLS	SAR: 1992 tables		SAR: Time-varying
	index	industry	Previous month Mcap	Previous day Mcap	Previous month Mcap
	(1)	(2)	(3)	(4)	(5)
			A. Point estimates		
β	–3.25***	–3.11***	–0.45***	–0.45***	–0.47***
	(0.71)	(0.13)	(0.12)	(0.12)	(0.13)
ρ			0.87***	0.87***	0.87***
			(0.01)	(0.01)	(0.01)
Constant	–0.13***	–0.11***	–0.01**	–0.01**	–0.02***
	(0.04)	(0.01)	(0.01)	(0.01)	(0.01)
Adj R^2 (%)	14.74	10.86	10.76	10.74	9.89
Observations	120	9,040	9,040	9,040	9,823
Log-L			–2,450	–2,447	–3,501
			B. Decomposition		
Direct effect 1			–0.72***	–0.72***	–0.68***
			(0.18)	(0.18)	(0.19)
Network effect 1			–2.65***	–2.64***	–2.83***
			(0.65)	(0.65)	(0.75)
Direct effect 2			–0.45***	–0.45***	–0.47***
			(0.12)	(0.12)	(0.13)
Network effect 2			–2.91***	–2.91***	–3.04***
			(0.72)	(0.72)	(0.81)

Table 1. Constant-coefficient OLS and SAR estimates.

Read.

- The aggregate stock-market response to a 1 percentage point surprise tightening is about -3.25%.
- The OLS industry response is about -3.11%.
- In the SAR specification, ρ is about 0.87 and statistically significant.
- Network effects account for roughly 80%–85% of the total response under the first two decompositions.

Economic message. A large part of what looks like an aggregate market reaction can be decomposed into higher-order propagation through input-output linkages.

5.3 Table 2: heterogeneous coefficients

Table 2
Response of industry returns to monetary policy shocks: Heterogenous coefficients and heteroscedastic error variance

	OLS	SAR: 1992 tables			
		Previous month Mcap	Previous day Mcap	Previous month Mcap	Previous day Mcap
		Homoscedastic errors		Heteroscedastic errors	
	(1)	(2)	(3)	(4)	(5)
		A. Point estimates			
β	–3.11***	–0.60***	–0.53***	–0.49***	–0.49***
	(0.13)	(0.18)	(0.16)	(0.15)	(0.15)
ρ		0.79***	0.82***	0.84***	0.84***
		(0.02)	(0.02)	(0.01)	(0.01)
Constant	–0.11***	–0.01	–0.01	–0.01	–0.01
	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)
Adj R^2 (%)	10.86	11.02	11.18	11.00	11.17
Observations	9,040	9,040	9,040	9,040	9,040
Log-L		–4,056	–4,056	–3,335	–3,313
		B. Decomposition			
Direct effect 1		–0.63***	–0.64***	–0.60***	–0.60***
		(0.19)	(0.19)	(0.18)	(0.18)
Network effect 1		–2.34***	–2.33***	–2.36***	–2.36***
		(0.67)	(0.66)	(0.67)	(0.67)
Direct effect 2		–0.60***	–0.53***	–0.49***	–0.49***
		(0.18)	(0.16)	(0.15)	(0.15)
Network effect 2		–2.38***	–2.44***	–2.47***	–2.47***
		(0.68)	(0.69)	(0.70)	(0.70)
Direct effect 3		–1.40***	–1.43***	–1.33***	–1.32***
		(0.42)	(0.42)	(0.39)	(0.39)
Network effect 3		–1.57***	–1.54***	–1.63***	–1.63***
		(0.48)	(0.46)	(0.48)	(0.48)

Table 2. Heterogeneous coefficients and heteroscedastic errors.

Read.

- Allowing industry-specific β_i and sectoral variation in ρ_i does not eliminate the network result.
- Average ρ remains around 0.79–0.84.
- Network effects remain about 80% of the total response in decompositions 1 and 2.
- The autarky comparison attributes about 55% of the total effect to network amplification.

Economic message. The baseline result is not an artifact of forcing all industries to have the same direct exposure to monetary policy.

5.4 Figure 2: empirical U.S. input-output network

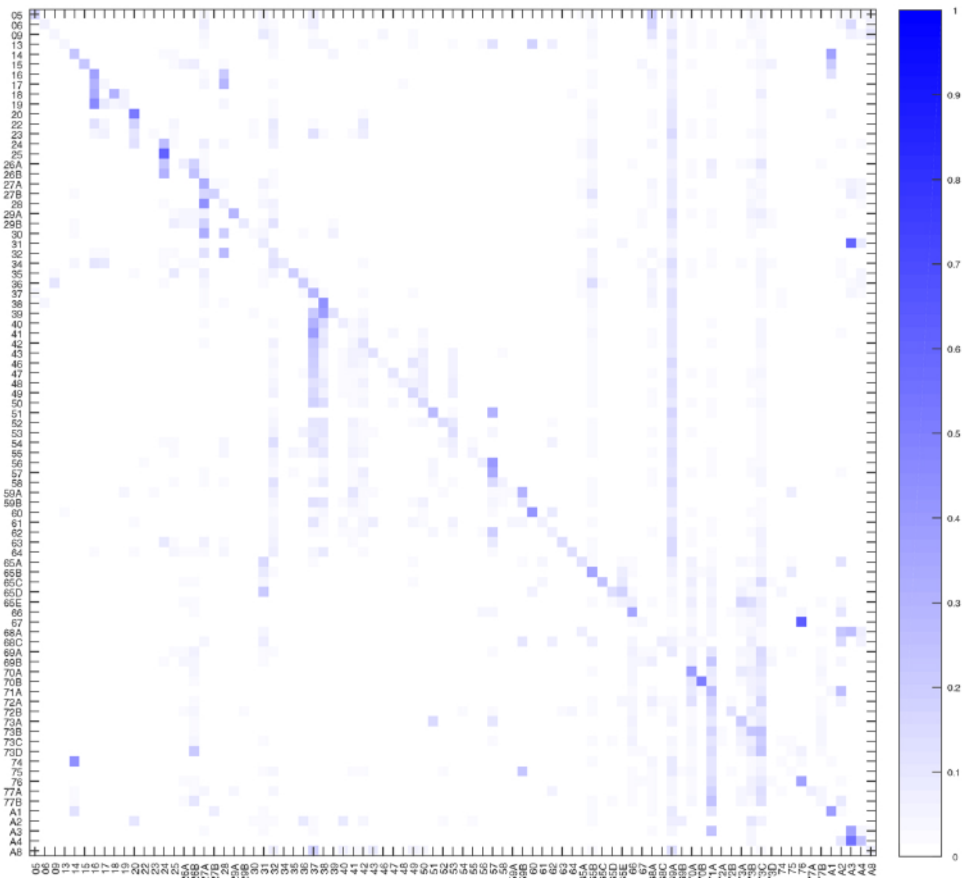


Figure 2. Heatmap of the empirical U.S. input-output matrix.

Read.

- The production network is sparse: many industry pairs have very small or zero links.
- Some industries are important suppliers to many other industries.
- There are visible diagonal and clustered structures, but the results are not driven only by diagonal entries.

Economic message. The particular structure of the U.S. production network is central. Network effects are not generated by arbitrary comovement.

5.5 Figures 3 and 4: placebo network simulations

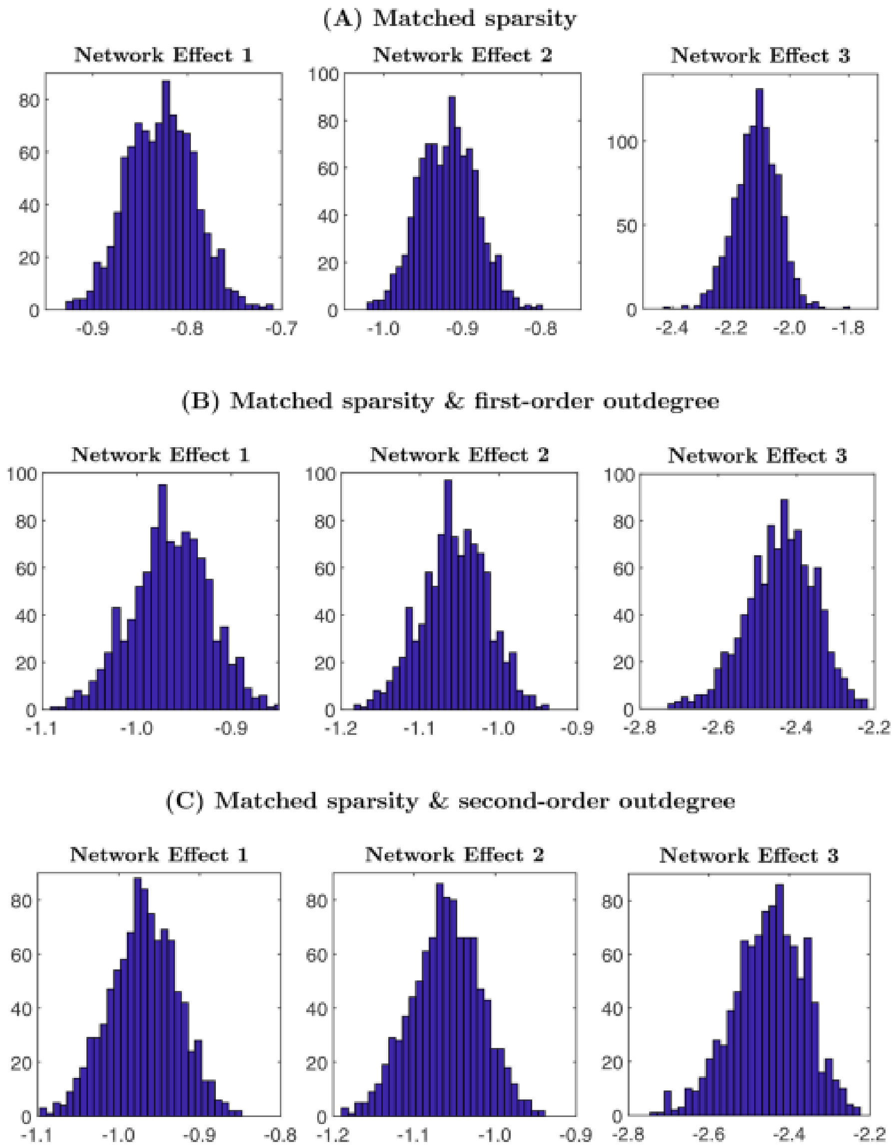


Figure 3. Simulated networks that match sparsity and outdegree distributions.

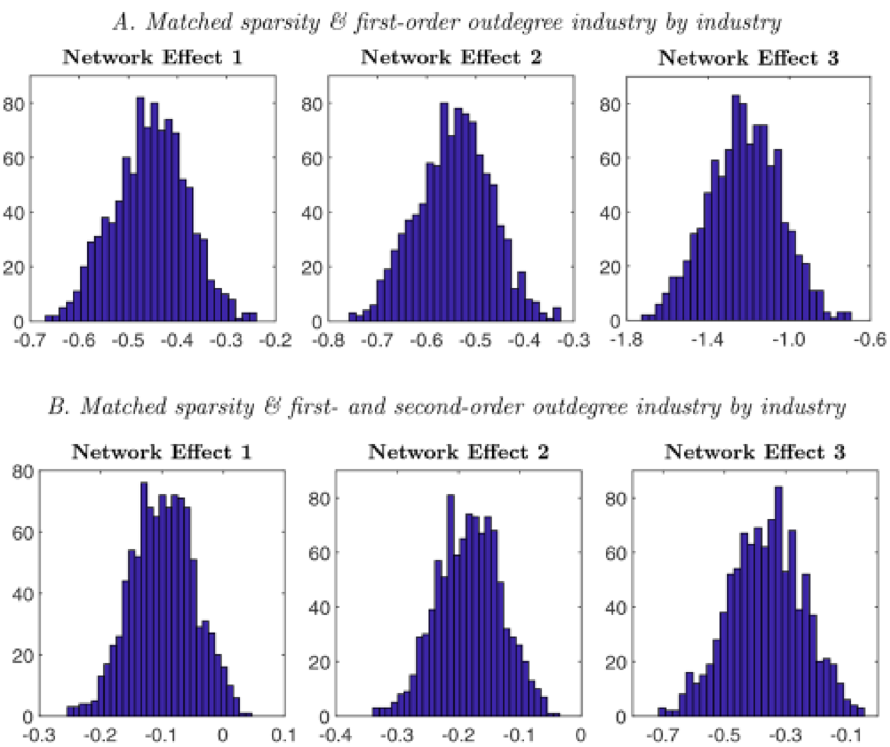


Figure 4. Simulated networks that match industry-by-industry network characteristics.

Read.

- Random networks with the same sparsity generate much smaller network effects than the empirical network.
- Matching the distribution of first- and second-order outdegrees is still not enough.
- Matching sparsity and outdegrees industry-by-industry gets much closer to the empirical estimates.

Economic message. What matters is not just the distribution of network statistics. It matters which industries occupy which positions in the actual U.S. production network.

5.6 Table 5: closeness to end-consumers

Table 5
Response of industry returns to monetary policy shocks by closeness to consumers

	Close to end-consumer (%) (1)	Far from end-consumer (%) (2)
Direct effect 1	88.57**	25.76***
Network effect 1	11.43**	74.24***
Direct effect 2	87.14**	24.45***
Network effect 2	12.86**	75.55***

Table 5. Direct and network effects by closeness to end-consumers.

Read.

- Industries close to end-consumers have direct effects around 87%–89%.
- Industries far from end-consumers have network effects around 74%–76%.
- The pattern is sharp and consistent across decompositions.

Economic message. Monetary policy is behaving like a demand shock: final-demand industries react directly, and upstream industries react through the network.

5.7 Table 6: external finance and price stickiness

Table 6
Response of industry returns to monetary policy shocks by external finance dependence and frequency of price adjustment

	A. External finance dependence	
	High (%) (1)	Low (%) (2)
Direct effect 1	16.97***	23.23***
Network effect 1	83.03***	76.77***
Direct effect 2	15.55***	22.44***
Network effect 2	84.45***	77.56***
Direct effect 3	41.25***	47.98***
Network effect 3	58.75***	52.02***
	B. Kaplan-Zingales Index	
	High (%)	Low (%)
Direct effect 1	25.22***	14.73**
Network effect 1	74.78***	85.27***
Direct effect 2	23.82***	13.98*
Network effect 2	76.18***	86.02***
Direct effect 3	53.54***	35.04***
Network effect 3	46.46***	64.96***
	C. Frequency price adjustment	
	High (%)	Low (%)
Direct effect 1	20.74***	19.80***
Network effect 1	79.26***	80.20***
Direct effect 2	19.53***	18.81***
Network effect 2	80.47***	81.19***
Direct effect 3	49.33***	40.40***

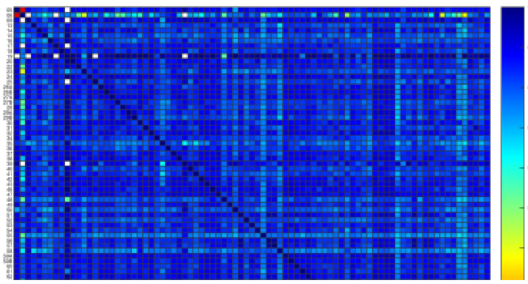
Table 6. Network shares by external finance dependence, KZ index, and price-adjustment frequency.

Read.

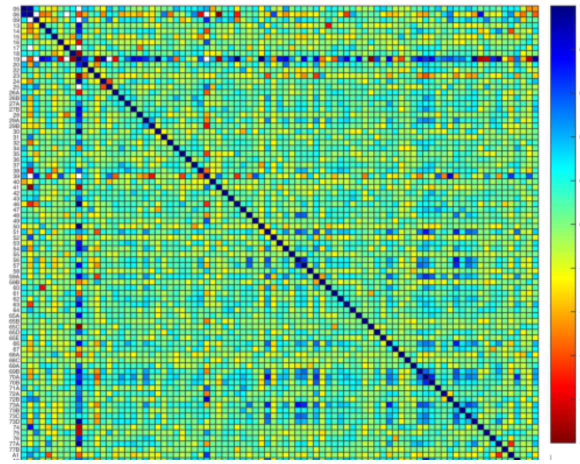
- High and low external-finance-dependence industries both show large network effects.
- High and low KZ-index industries also both show large network effects.
- High and low price-adjustment-frequency industries have similar network shares.

Economic message. The main mechanism is not primarily financial constraints or heterogeneous price rigidity. The demand-propagation channel remains central.

5.8 Figures 5 and 6: cross-industry correlation patterns



FOMC event-window correlations



Correlations after removing network effects

Figures 5 and 6. Industry-return correlations before and after removing network effects.

Read.

- Industry returns are highly correlated around FOMC announcements.
- After removing estimated network effects, correlations become much weaker.
- The remaining correlation after removing network effects is small relative to raw FOMC correlations.

Economic message. Production networks explain much of the cross-industry comovement generated by monetary policy news.

5.9 Table 7: realized fundamentals

Table 7
Response of industry cash flow fundamentals to monetary policy shocks

Horizon	0	1	2	3	4	5	6	7	8
					A. Value-weighted sales				
Direct effect 1	1.26*	1.71*	2.22**	2.64**	2.77**	2.61*	2.50	2.51	2.71*
	(0.74)	(0.89)	(0.98)	(1.11)	(1.27)	(1.42)	(1.55)	(1.61)	(1.61)
Network effect 1	1.92*	2.48**	3.13**	3.52**	3.61**	3.50**	3.29*	3.16*	3.21*
	(1.09)	(1.26)	(1.35)	(1.44)	(1.53)	(1.62)	(1.73)	(1.78)	(1.81)
					B. Equally-weighted sales				
Direct effect 1	1.10*	1.44**	1.74**	1.97**	1.94**	1.74*	1.56	1.43	1.39
	(0.61)	(0.69)	(0.77)	(0.88)	(0.97)	(1.04)	(1.06)	(1.04)	(0.99)
Network effect 1	2.27	3.05*	3.69*	4.03*	4.08*	3.87	3.70	3.40	3.21
	(1.48)	(1.72)	(1.91)	(2.13)	(2.30)	(2.41)	(2.46)	(2.47)	(2.45)
					C. Value-weighted operating income				
Direct effect 1	0.34**	0.42**	0.51***	0.54**	0.52**	0.46*	0.39	0.40	0.43*
	(0.15)	(0.18)	(0.19)	(0.22)	(0.24)	(0.26)	(0.26)	(0.25)	(0.26)
Network effect 1	0.61**	0.75***	0.88***	0.92***	0.90**	0.83**	0.78**	0.75**	0.77**
	(0.25)	(0.28)	(0.31)	(0.34)	(0.35)	(0.37)	(0.36)	(0.35)	(0.36)
					D. Equally-weighted operating income				
Direct effect 1	0.30**	0.36***	0.41***	0.43***	0.40**	0.34**	0.25*	0.21*	0.19*
	(0.12)	(0.13)	(0.14)	(0.16)	(0.16)	(0.16)	(0.14)	(0.12)	(0.10)
Network effect 1	0.71***	0.86***	0.95***	0.97***	0.93**	0.83**	0.76**	0.68**	0.62**
	(0.27)	(0.29)	(0.33)	(0.37)	(0.39)	(0.40)	(0.36)	(0.33)	(0.30)

Table 7. Network effects in sales and operating income.

Read.

- Network effects appear in value-weighted and equal-weighted sales.
- Network effects also appear in value-weighted and equal-weighted operating income.
- The network component is about 60% of the immediate response and grows over several quarters.

Economic message. The stock-market evidence is not only a valuation artifact. Realized cash-flow fundamentals also show network propagation.

5.10 Table 8: dynamic model simulation

Table 8
Direct and network effects from simulated data

Model parameters: Benchmark	Model parameters: Variation	β	ρ	% network 1	% network 2	% network 3
	Benchmark	−0.73	0.78	67.00	77.16	51.81
		(0.03)	(0.01)			
$\eta=0.1$	$\eta=0.2$	−0.70	0.79	68.18	78.09	54.04
		(0.01)	(0.01)			
$a=0.85$	$a=0.7$	−0.83	0.74	63.21	74.03	44.27
		(0.04)	(0.01)			
$\phi=-0.5$	$\phi=-0.4$	−0.83	0.74	63.04	73.88	43.92
		(0.04)	(0.01)			
$m/w=1$	$m/w=2$	−0.71	0.78	67.48	77.54	52.74
		(0.02)	(0.01)			
$\rho_M=0.5$	$\rho_M=0.75$	−1.19	0.78	67.25	77.36	52.29
		(0.04)	(0.01)			
$\sigma_M=0.01$	$\sigma_M=0.02$	−0.72	0.78	67.09	77.23	51.98
		(0.03)	(0.01)			
$\psi=0.2$	$\psi=0.4$	−0.72	0.78	67.23	77.35	52.26
		(0.02)	(0.01)			
BEA W	$W=[1/N]$	−0.28	0.63	74.46	75.72	1.37
		(0.03)	(0.02)			

Table 8. Direct and network effects from simulated dynamic-model data.

Read.

- The benchmark simulated model generates ρ around 0.78.
- Network shares are about 67%, 77%, and 52% under the three decompositions.
- The result is stable across alternative calibrations.
- A dense equal-weight network produces almost no amplification under decomposition 3.

Economic message. A dynamic production-network model can generate network shares close to the empirical estimates, reinforcing the interpretation of the SAR evidence.

6 Short summary

- The paper studies how monetary policy shocks propagate through production networks.
- It uses FOMC event-window industry returns, federal funds futures surprises, and BEA input-output tables.
- A simple model predicts a spatial autoregression for industry returns.
- The estimated network parameter is large and statistically significant.
- Network effects account for roughly 55%–85% of the overall stock-market response.
- Industries close to end-consumers respond mostly directly; upstream industries respond mostly through the network.

- Random placebo networks cannot easily reproduce the empirical network effects.
- Realized fundamentals show similar network propagation.
- Dynamic-model simulations generate network effects close to the empirical magnitudes.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

Ozdagli and Weber show that production networks are a major channel for monetary policy transmission. Industry stock returns around FOMC announcements do not merely reflect direct exposure to monetary policy shocks; they also reflect higher-order demand propagation through input-output linkages.

7.2 Economic intuition

A monetary expansion raises final demand. Consumer-facing firms increase production and therefore demand more intermediate inputs. Their suppliers then increase production and demand inputs from their own suppliers. Because the network is interconnected and cyclic, these effects continue across many orders and can feed back to initially affected sectors.

7.3 Takeaway

The paper's main takeaway is that macro-finance models should not treat industries as isolated units when studying aggregate policy shocks. The structure of the production network determines how monetary policy moves from final demand to upstream sectors and helps explain why aggregate stock-market responses can be so large.